

Open Problems on Matroid Theory and Submodular Optimization

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Instructor note: GPT 6 Astra is used to generate this note based on topics I selected (with much bias based on what I know). The sources of all the open problems are clearly listed, and you may find more background knowledge from the cited papers. Feel free to suggest your own topic and let me know. You are encouraged to try, but do not have to make progress on any of them. Making small observations and solving special cases are very welcome. Please submit a 5-10 page report explaining the literature, give expositions of papers you read, and/or discuss progress you made. AI tools are allowed as you explore the problems, but the final report must be written by yourself.

Abstract

This note collects several open problems from two themes: matroid theory and submodular optimization. The topics include exact and group-constrained bases, bounded-subdeterminant integer programming, fast packing and covering, basis decompositions, matroid exchange, discrepancy rounding and weighted equitability, submodular maximization and partitioning, and contention resolution. Each section gives the relevant background and known results, with citations, followed by a precise open question.

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Minimal background

A *matroid* $M = (E, \mathcal{I})$ consists of a finite ground set E and a family $\mathcal{I}$ of subsets called independent sets. Besides being closed under taking subsets, the independent sets satisfy an exchange rule: if $I, J \in \mathcal{I}$ and $|I| < |J|$, then some element of $J \setminus I$ can be added to I while preserving independence. The maximal independent sets are the *bases*; all have the same size, the rank $r(M)$. Linear independence of vectors and acyclic edge sets in a graph are the two guiding examples. In the latter case, the bases are spanning trees.

The characteristic vector of $B \subseteq E$ is denoted by $\mathbf{1}_B$, and the base polytope is

$$P_B(M) = \text{conv}\{\mathbf{1}_B : B \text{ is a basis of } M\}.$$

An *independence oracle* answers whether a queried set is independent. A *represented matroid* is given more concretely by a matrix whose columns are indexed by E . This distinction matters: algebraic algorithms can exploit a representation in ways that are impossible in the oracle model.

Part I

Matroid Theory

1 A matroidal Kneser theorem for group-labeled bases

Background. Label each element of a matroid by an element of a finite abelian group Γ . The label of a basis is the sum of its element labels. For instance, with $\Gamma = \mathbb{Z}_2$, the label records the parity of the number of marked elements in the basis. As bases exchange elements, their attainable sums tend to spread throughout the group. The only evident obstruction is periodicity: the attainable set may be a union of cosets of a nontrivial subgroup.

For a labeling $\psi : E \rightarrow \Gamma$, set

$$S = \psi(\mathcal{B}(M)) := \left\{ \sum_{e \in B} \psi(e) : B \in \mathcal{B}(M) \right\}, \quad H = \text{stab}(S) := \{h \in \Gamma : S + h = S\}.$$

For a coset $Q \in \Gamma/H$, let $E_Q = \{e \in E : \psi(e) \in Q\}$. The subgroup H measures the periodicity of the attainable basis sums.

Kneser’s addition theorem says, roughly, that a sumset must grow unless it has subgroup symmetry. Here the rank contribution $r_M(E_Q)$ measures how many independent choices can be drawn from each label coset, while H captures the same symmetry obstruction. Schrijver and Seymour proved the prime-order case [63]. DeVos, Goddyn, and Mohar proved the conjecture when $|\Gamma|$ is a product of two distinct primes and when Γ is cyclic of prime-power order; the conjecture also holds for uniform matroids [26]. It remains open for general matroids and finite abelian groups. Its importance is now primarily structural: exact group-labeled basis feasibility is fixed-parameter tractable by other methods, but the conjecture would explain the sharp exchange and proximity phenomena behind that tractability [49, 33].

Open Question 1 (Schrijver–Seymour conjecture). For every matroid M and every labeling into a finite abelian group, is

$$|S| \geq |H| \left(1 - r(M) + \sum_{Q \in \Gamma/H} r_M(E_Q) \right)?$$

2 A derandomization hierarchy for exact matroid intersection

Background. A set may be required to be a basis in two matroids simultaneously. For two partition matroids this recovers perfect matchings in a bipartite graph: one matroid enforces one edge at every left vertex and the other enforces one edge at every right vertex. This gives two elementary benchmark problems. In *Exact Bipartite Matching*, the edges are red or blue and one asks for a perfect matching with exactly t red edges. In the congruency-constrained version, one prescribes

$$|M \cap R| \equiv a \pmod{m}.$$

The parity case $m = 2$ is deterministically solvable even for general graphs [34]; fixed moduli $m \geq 3$ therefore form a natural next rung. When the modulus is part of the input, the two red-count decision problems are polynomial-time Turing equivalent. On a bipartite graph with n vertices on each side, exact matching is the congruency problem with $m = n + 1$, while a congruency instance can be solved using at most $n + 1$ exact-matching queries, one for each feasible count in the required residue class.

Randomized polynomial-time algorithms have long been known for Exact Matching and exact matroid problems [54, 17]. A recent unrefered preprint claims a deterministic algorithm using $O(n^6)$ arithmetic operations for Exact Bipartite Matching [28]; if correct, the preceding reduction also settles the red-count congruency version. Thus these two problems now serve as verification and simplification benchmarks rather than unqualified open problems.

The broader benchmark replaces the two partition matroids by represented linear matroids. Adding a 0–1 label to each element and prescribing its total asks for a common basis containing exactly k marked elements.

Suppose two rank- r matroids are represented over the same field by matrices A_1, A_2 with columns indexed by E . Cauchy–Binet expands

$$\det\left(A_1 \operatorname{diag}(x_e z^{\ell(e)} : e \in E) A_2^T\right) = \sum_{B \in \binom{E}{r}} \det(A_1[B]) \det(A_2[B]) \left(\prod_{e \in B} x_e \right) z^{\ell(B)}.$$

A term is present precisely when B is a common basis. Random evaluation of the x_e 's, followed by polynomial identity testing, yields randomized polynomial-time algorithms for exact-weight and, more generally, finite-group label constraints [17, 43]. Deterministically certifying that the relevant coefficient is not the zero polynomial is the missing step.

The representation assumption is essential: related group-labeled common-basis problems require exponentially many queries for arbitrary independence-oracle matroids [43]. Deterministic algorithms are known for special algebraic structures, including Pfaffian pairs [51]. Thus the problem is a concrete derandomization question about a structured determinant polynomial, not merely a request to improve a generic oracle algorithm.

Open Question 2 (From exact matching to exact linear common bases). Obtain deterministic algorithms or independently verified proofs along the following hierarchy:

- (a) give a direct deterministic polynomial-time algorithm for red-count congruency-constrained bipartite perfect matching for each fixed modulus $m \geq 3$;
- (b) independently verify, repair, or replace the claimed deterministic algorithm for Exact Bipartite Matching; and
- (c) given binary-encoded rational representations of two rank- r matroids on E , labels $\ell : E \rightarrow \{0, 1\}$, and $k \in \{0, \dots, r\}$, decide whether they have a common basis B with $\sum_{e \in B} \ell(e) = k$, and find one when it exists.

In light of [28], parts (a) and (b) should be read as requests for independent verification or a simpler deterministic proof; part (c) is the principal open problem.

3 The Δ -modular integer-programming conjecture

Background. Total unimodularity explains why many linear programs arising in combinatorial optimization have integral optima. A far-reaching proposed extension controls only the largest full-dimensional subdeterminant. For a full-column-rank matrix $A \in \mathbb{Z}^{m \times n}$, define

$$\Delta(A) = \max\{|\det A_I| : I \subseteq [m], |I| = n\},$$

where A_I is the square submatrix induced by the rows in I . The matrix is called Δ -modular when $\Delta(A) \leq \Delta$. The associated optimization problem is

$$\min\{c^\top x : Ax \leq b, x \in \mathbb{Z}^n\},$$

with A, b, c given in binary.

The folklore Δ -modular IP conjecture asks whether this problem is polynomial-time solvable for every fixed Δ . The case $\Delta = 1$ follows from linear programming, and the bimodular case $\Delta = 2$ has a strongly polynomial algorithm [7]; the conjecture remains open already for $\Delta = 3$. Important restricted cases are known. For example, fixed- Δ instances with at most two nonzero entries in every row or every column are strongly polynomial-time solvable [38]. A matrix is *strictly* Δ -modular when each full-dimensional minor lies in $\{-\Delta, 0, \Delta\}$; even this subclass captures totally unimodular problems with congruency constraints, including congruency-constrained matching. Randomized strongly polynomial feasibility algorithms are known for the strict subclass when $\Delta \leq 4$ [55].

Open Question 3 (Δ -modular IP conjecture). For every fixed $\Delta \geq 3$, can

$$\min\{c^\top x : Ax \leq b, x \in \mathbb{Z}^n\}$$

be solved in time polynomial in the binary input length whenever $\Delta(A) \leq \Delta$? In particular, can the conjecture be proved for $\Delta = 3$, and can the resulting algorithm be made strongly polynomial? A first unresolved milestone is optimization for strictly 3-modular matrices.

4 Fast algorithms for matroid packing and covering

Background. Let $M = (E, \mathcal{I})$ have $n = |E|$ elements and rank $r > 0$, and assume that M is loopless when discussing coverings. Its *base-packing number* $p(M)$ is the maximum number of pairwise disjoint bases. Its *chromatic number*, or matroid arboricity, $\chi(M)$ is the minimum number of independent sets that partition E . The latter is also the minimum number of bases whose union covers E : a cover can be made disjoint by assigning each element to one set containing it, while every independent set can be extended to a basis.

Edmonds's packing and partition theorems give the exact min-max formulas

$$p(M) = \left\lfloor \min_{\substack{X \subseteq E \\ \text{rk}(X) < r}} \frac{|E \setminus X|}{r - \text{rk}(X)} \right\rfloor, \quad \chi(M) = \left\lceil \max_{\emptyset \neq X \subseteq E} \frac{|X|}{\text{rk}(X)} \right\rceil \quad [30, 31].$$

For an integer k , let

$$U_k(M) = \max \left\{ \left| \bigcup_{i=1}^k I_i \right| : I_1, \dots, I_k \in \mathcal{I} \right\}.$$

The matroid-union theorem states that

$$U_k(M) = \min_{X \subseteq E} (|E \setminus X| + k \text{rk}(X)).$$

Consequently, $p(M) \geq k$ exactly when $U_k(M) = kr$, whereas $\chi(M) \leq k$ exactly when $U_k(M) = n$.

Quanrud's deterministic push-relabel and augmentation algorithms compute $U_k(M)$ in

$$O\left(n + U_k(M) \sqrt{n' \log(kr)}\right) \quad \text{independence queries,} \quad n' = \min\{n + U_k(M) \log r, rk \log(kr)\} [61].$$

Thus, binary search over k computes both $p(M)$ and $\chi(M)$, together with decompositions and min-max certificates, in $\tilde{O}(n^{3/2})$ queries. The same framework gives deterministic $(1 \pm \varepsilon)$ decision procedures in $\tilde{O}(n/\varepsilon)$ queries, leaving a substantial gap between approximate and exact computation.

The graphic special cases make this gap concrete. Let $G = (V, E)$ be a loopless graph with $N = |V|$ and $m = |E|$. The covering number of its graphic matroid is the graph arboricity

$$a(G) = \max_{\substack{S \subseteq V \\ |S| \geq 2}} \left\lceil \frac{|E(S)|}{|S| - 1} \right\rceil.$$

If G is connected, its base-packing number is the maximum number $\tau(G)$ of edge-disjoint spanning trees, and

$$\tau(G) = \min_{\substack{\mathcal{P} \text{ a partition of } V \\ |\mathcal{P}| \geq 2}} \left\lceil \frac{|\delta(\mathcal{P})|}{|\mathcal{P}| - 1} \right\rceil = \lfloor \sigma(G) \rfloor,$$

where $\sigma(G)$ is the fractional graph strength [57, 56, 65]. Cen et al. compute exact arboricity, even with polynomially bounded integral edge weights, in randomized $\sqrt{N} m^{1+o(1)}$ time; their randomized $mN^{1+o(1)}$ -time cut-hierarchy algorithm also computes $\sigma(G)$ [18]. For unweighted multigraphs, deterministic matroid-union methods compute a maximum integral spanning-tree packing in $\tilde{O}(m^{3/2})$ time [61]. If k is fixed, a newer deterministic algorithm finds a maximum union of k forests, and hence decides and constructs k edge-disjoint spanning trees when they exist, in

$$O\left(k^3 \min\{kN, m\} \log^2 N + k m^{1+o(1)}\right)$$

time [5].

Open Question 4 (Near-linear matroid packing and covering). Can $p(M)$ and $\chi(M)$, together with optimal decompositions and min–max certificates, be computed with $\tilde{O}(n)$ independence queries? More generally, can one improve the $\tilde{O}(n^{3/2})$ worst-case bound, or prove a superlinear lower bound in the standard independence-oracle model? For graphic matroids, can $a(G)$ and $\tau(G)$ be computed deterministically in $m^{1+o(1)}$ time when $k = \tau(G)$ is not fixed, and can exact fractional strength or the entire cut hierarchy be computed in $m^{1+o(1)}$ time?

5 Woodall’s conjecture on packing dijoins

Background. In a digraph $D = (V, A)$, a *dicut* is a nonempty directed cut $\delta^+(U)$ for which no arc enters U , that is, $\delta^-(U) = \emptyset$. A *dijoin* is a set of arcs meeting every dicut. Think of each dicut as a one-way bottleneck and of a dijoin as a set of arcs that places at least one guard on every such bottleneck.

Let

$$\tau(D) = \min\{|C| : C \text{ is a dicut of } D\}, \quad \nu(D) = \max\{t : D \text{ has } t \text{ pairwise arc-disjoint dijoins}\}.$$

Every dijoin must use an arc from a smallest dicut, so the elementary bound $\nu(D) \leq \tau(D)$ is immediate.

Woodall posed the conjecture in 1976 [67]. It holds for $\tau \leq 2$ but remains open already for $\tau = 3$, even for planar digraphs [23]. A polynomial-time construction of at least $\lfloor \tau/6 \rfloor$ disjoint dijoins is known [24]. The weighted analogue conjectured by Edmonds and Giles is false in general, so it should not be conflated with Woodall’s unweighted statement [1]; it does hold when the underlying undirected graph is chordal [25]. Woodall’s conjecture is also a special case of packing common bases of two matroids [39, 11], placing it inside the broader framework of Question 6.

Open Question 5 (Woodall’s dijoin conjecture). Does every digraph with at least one dicut satisfy

$$\nu(D) = \tau(D)?$$

Equivalently, can every digraph whose smallest dicut has size τ pack τ pairwise arc-disjoint dijoins?

6 A constant integrality gap for packing common bases

Background. For one matroid, Edmonds’s base-packing theorem gives an exact min–max description of how many disjoint bases can be packed [30]. Requiring every chosen set to be a basis of *two* matroids changes the problem dramatically. One common basis is computable by weighted matroid

intersection, yet packing several common bases already contains difficult problems such as Woodall's and Rota's conjectures [11].

Let M_1 and M_2 be rank- r matroids on the same ground set, let $\mathcal{B}_{12} = \mathcal{B}(M_1) \cap \mathcal{B}(M_2)$, and define

$$\begin{aligned}\nu(M_1, M_2) &= \max \left\{ \sum_{B \in \mathcal{B}_{12}} y_B : \sum_{B \ni e} y_B \leq 1 \ (e \in E), \ y_B \in \mathbb{Z}_{\geq 0} \right\}, \\ \nu^*(M_1, M_2) &= \max \left\{ \sum_{B \in \mathcal{B}_{12}} y_B : \sum_{B \ni e} y_B \leq 1 \ (e \in E), \ y_B \geq 0 \right\}.\end{aligned}$$

The first value is the maximum number of pairwise disjoint common bases; the second permits fractional copies of bases. Clearly $\nu \leq \nu^*$.

The fractional problem is polynomial-time solvable through weighted matroid intersection [32], whereas deciding exact decomposability into common bases is NP-hard in explicitly represented special cases and has exponential oracle complexity even for two bases [11]. Thus equality is too much to expect in full generality, while a constant ratio would still be a robust approximate min-max theorem. The constant-factor packing of dijoins in Question 5 supplies evidence in a highly structured special case and motivates asking for a universal bound. A short maximality argument gives only the rank-dependent estimate $\nu \geq \nu^*/r$: the union of a maximal packing has $r\nu$ elements and meets every common basis. Already K_4 gives a factor-two gap: intersect its graphic matroid with the partition matroid whose three blocks are the three pairs of opposite edges. The four common bases are the four spanning stars; any two share an edge, while assigning weight 1/2 to each star is a fractional packing of value 2. Thus $\nu = 1$, $\nu^* = 2$, and any universal c can be at most 1/2 [41].

Open Question 6 (Constant-gap common-base packing). Is there a universal constant $c > 0$ such that, for every pair of matroids on a common ground set and of the same rank,

$$\nu(M_1, M_2) \geq c \nu^*(M_1, M_2)?$$

Can such a guarantee be made constructive in natural representation models?

7 Sampling decompositions into matroid bases

Background. There is an important distinction between sampling a union of bases and sampling its decomposition. Suppose that M has rank r and that a set $S \subseteq E$ of size kr can be partitioned into k bases of M . Then S is a basis of the k -fold matroid union $M^{\vee k}$. Thus sampling S from all such unions reduces to ordinary matroid-basis sampling: matroid partition provides an independence oracle for $M^{\vee k}$, and the down-up walk samples a basis of any oracle matroid approximately uniformly [31, 4].

The unresolved problem begins after the union S has been fixed. Let

$$\mathcal{D}_k(M, S) = \left\{ (B_1, \dots, B_k) \in \mathcal{B}(M)^k : S = B_1 \dot{\cup} \dots \dot{\cup} B_k \right\}$$

be the set of ordered decompositions of S into bases. Matroid partition can decide whether this set is nonempty and produce one member [31]; sampling nearly uniformly from all its members is a different problem.

A symmetric exchange between B_i and B_j preserves their union. White conjectured that any two ordered basis tuples with the same multiset union are connected by such exchanges [66]. The

conjecture remains open in general, although it is now known for paving matroids [68]. Even connectivity would not by itself prove that the natural exchange walk mixes rapidly.

When $k = 2$, write $N = M|S$. The map $(B_1, B_2) \mapsto B_1$ identifies $\mathcal{D}_2(M, S)$ with the common bases of N and its dual N^* . Sampling a uniformly random common basis of a matroid and its dual was posed at the 2024 Cargèse Workshop as the first test case [8].

Open Question 7 (Sample a decomposition with a prescribed union). Given an independence-oracle matroid M , an integer $k \geq 2$, a set S with $\mathcal{D}_k(M, S) \neq \emptyset$, and $\varepsilon > 0$, can one sample in time polynomial in $|S|$, k , and $\log(1/\varepsilon)$ from a distribution whose total-variation distance from the uniform distribution on $\mathcal{D}_k(M, S)$ is at most ε ? In particular, can one solve the case $k = 2$, or prove rapid mixing of a natural lazy symmetric-exchange walk?

8 Rota’s basis conjecture

Background. Place r bases of a rank- r matroid into the rows of an $r \times r$ array. One may permute entries within each row. Rota’s conjecture asks whether the array can always be rearranged so that every column is also a basis. If an element occurs in more than one row, its occurrences may be treated as distinct parallel copies.

Call a basis *transversal* (or rainbow) if it takes exactly one element from each original B_i . The conjecture asks for r pairwise disjoint transversal bases covering all r^2 occurrences. It remains open even for vector matroids [44, 41]. A major 2025 advance proves that one can always pack $(1 - o(1))r$ disjoint transversal bases and cover all elements with $(1 + o(1))r$ transversal bases [53]. A 2026 randomized polynomial-time result constructs the corresponding asymptotic covering when the rank is sufficiently large [6]. These asymptotically sharp results do not yet reach the exact target of r disjoint columns.

Open Question 8 (Rota’s basis conjecture). Let $B_1, \dots, B_r$ be r bases of a rank- r matroid. Can each row be ordered as $B_i = (b_{i1}, \dots, b_{ir})$ so that

$$\{b_{1j}, b_{2j}, \dots, b_{rj}\}$$

is a basis for every $j \in [r]$?

9 Gabow’s serial symmetric exchange conjecture

Background. The symmetric basis-exchange theorem says that, for two bases, one can swap a suitable element of the first with a suitable element of the second and keep both sets as bases. Gabow asks whether all elements can be paired and swapped *serially*: after every prefix of swaps, both evolving sets must still be bases. Locally valid exchanges need not obviously fit into one globally valid order.

On $A \cup B$, the conjecture is equivalent to finding a cyclic ordering in which A and B are consecutive blocks and every r consecutive elements form a basis. The conjecture is known for several important classes, including graphic, regular, strongly base-orderable, sparse paving, and split matroids, but is open for arbitrary matroids [40, 45, 9]. It illustrates why elementary exchange proofs can carry algorithmic information that a nonconstructive existence proof may conceal.

Open Question 9 (Gabow’s serial symmetric exchange conjecture). Let A and B be disjoint bases of a rank- r matroid. Do there exist orderings $A = (a_1, \dots, a_r)$ and $B = (b_1, \dots, b_r)$ such that, for every $j \in \{0, \dots, r\}$, both

$$(A \setminus \{a_1, \dots, a_j\}) \cup \{b_1, \dots, b_j\} \quad \text{and} \quad (B \setminus \{b_1, \dots, b_j\}) \cup \{a_1, \dots, a_j\}$$

are bases? If so, can the two orderings be found with polynomially many independence-oracle queries?

10 Constant prefix discrepancy for a matroid basis

Background. Let the elements arrive in the fixed order $1, 2, \dots, n$. A fractional basis $x \in P_B(M)$ chooses a fraction x_i of each element. We want to round x to a genuine basis without letting the rounded and fractional weighted totals drift far apart at *any time* t . Controlling only the final total is much easier than simultaneously controlling all n prefixes.

The constraints are nested, but an early element participates in almost every later prefix, so ordinary column-sparse discrepancy bounds do not directly give a constant. The current general guarantee is $C = O(\log n)$, while $C = 1$ is known for binary weights and for uniform matroids with arbitrary nonnegative weights [10]. A constant would imply, up to constant factors, the prefix-constrained generalized-assignment conjecture and would feed into approximation algorithms for maximum flow-time scheduling. This is a particularly direct example of discrepancy theory producing a new angle on matroid optimization. The same work proves that any discrepancy guarantee C can be made cost-preserving: for every cost vector c , one can ensure $c(B) \leq c^\top x$ while increasing the discrepancy bound to at most $2C$ [10]. Thus cost preservation creates no separate asymptotic obstacle once a constant discrepancy bound is known.

Open Question 10 (Prefix-constrained matroid basis conjecture). Given a matroid M on the ordered ground set $[n]$, a fractional basis $x \in P_B(M)$, and nonnegative weights $w \in \mathbb{R}_{\geq 0}^n$, is there a universal constant C and a basis B satisfying

$$\left| \sum_{i=1}^t w_i (\mathbf{1}_B(i) - x_i) \right| \leq C \|w\|_\infty \quad \text{for every } t \in [n]?$$

If so, what is the smallest possible universal constant, and can such a basis be found with polynomially many independence-oracle queries?

11 Sparse weighted equitability of matroid bases

Background. Suppose that the ground set E of a matroid M can be partitioned into k bases. For one nonnegative weight vector w , weighted matroid equitability guarantees a partition $E = B_1 \sqcup \dots \sqcup B_k$ satisfying

$$\left| w(B_j) - \frac{w(E)}{k} \right| \leq \left(1 - \frac{1}{k} \right) \max_{e \in E} w_e \quad (j \in [k]) [10].$$

A natural extension asks to balance many weight functions simultaneously. Represent them by the rows of a nonnegative matrix $W \in \mathbb{R}_{\geq 0}^{m \times E}$, write $w_i(S) = \sum_{e \in S} W_{i,e}$, and let Δ be the maximum number of nonzero entries in a column of W .

There is already a one-basis theorem. The assumed decomposition implies $\mathbf{1}_E/k \in P_B(M)$, and discrepancy rounding produces a basis B such that

$$\left\| W \mathbf{1}_B - \frac{1}{k} W \mathbf{1}_E \right\|_\infty \leq 2\Delta \max_{u \in [m], e \in E} W_{u,e}.$$

However, $E \setminus B$ need not be partitionable into $k - 1$ bases, so this result cannot simply be applied recursively to obtain a balanced full decomposition [10].

The multi-constraint question strengthens Conjecture 1.5 of *Matroids Are Equitable* [3]. For pairwise disjoint sets $S_1, \dots, S_\ell$, set $W_{i,e} = \mathbf{1}\{e \in S_i\}$. The resulting matrix has column sparsity one, and balancing its rows means balancing all the cardinalities $|B_j \cap S_i|$. The older conjecture allows the discrepancy bound to depend on ℓ ; the sparse weighted formulation below asks for a universal value $g(1)$ independent of the number of sets, and also permits weighted, overlapping supports when $\Delta > 1$.

Open Question 11 (Sparse multi-constraint weighted equitability). Does there exist a function $g: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ such that, for every matroid $M = ([n], \mathcal{I})$ whose ground set is partitionable into k bases and every nonnegative matrix $W \in \mathbb{R}_{\geq 0}^{m \times n}$ with at most Δ nonzeros per column, there is a partition $[n] = B_1 \sqcup \dots \sqcup B_k$ into bases satisfying

$$\left| w_i(B_j) - \frac{w_i([n])}{k} \right| \leq g(\Delta) \max_{u \in [m], v \in [n]} W_{u,v} \quad \text{for every } i \in [m] \text{ and } j \in [k]?$$

Part II

Submodular Optimization

12 Nonmonotone submodular maximization under cardinality

Background. A set function $f: 2^E \rightarrow \mathbb{R}_{\geq 0}$ is submodular if it has diminishing marginal returns. It is *nonmonotone* when adding an element can decrease its value; cut functions are a basic example. Given value-oracle access to f , the cardinality-constrained problem is

$$\text{OPT} = \max\{f(S) : S \subseteq E, |S| \leq r\}.$$

The sets of size at most r form a uniform matroid. Thus this is both the simplest nontrivial matroid constraint and a benchmark for all down-closed constraints. The qualifier “at most” matters: for a nonmonotone function, an optimal set need not have size exactly r .

For $x \in [0, 1]^E$, let $F(x) = \mathbb{E}[f(R(x))]$, where $R(x)$ contains each element e independently with probability x_e . This multilinear relaxation is especially revealing here: over a matroid polytope, pipage or swap rounding can convert x into an independent set without decreasing the expected value. Consequently, the main unresolved loss is in finding a good fractional point, not in rounding it.

Buchbinder and Feldman give a randomized polynomial-time 0.401-approximation for every matroid constraint, and hence for a cardinality constraint [14]. Conversely, Qi constructs cardinality-constrained instances for which a randomized 0.478-approximation requires exponentially many value queries [60, Theorem 4.1]. The polynomial-query threshold is therefore commonly summarized as lying between 0.401 and 0.478; the latter number is an oracle barrier, not a matching hardness threshold.

Open Question 12 (Nonmonotone cardinality-constrained maximization). Determine the optimal approximation factor achievable with polynomially many value-oracle queries for maximizing a nonnegative, nonmonotone submodular function subject to $|S| \leq r$. Can the 0.401 guarantee be improved by a fixed constant, or can the 0.478 oracle barrier be lowered toward the algorithmic guarantee?

13 Optimal one-pass streaming under a matroid

Background. A submodular function models diminishing returns: the benefit of adding an element can only decrease as the chosen set grows. Maximum coverage is the standard example. Suppose that the elements arrive one at a time in uniformly random order and that the algorithm may retain only about as many elements as the rank of a matroid. It must decide what information to keep without ever storing the full instance.

Let $M = (E, \mathcal{I})$ be a rank- r matroid and let $f : 2^E \rightarrow \mathbb{R}_{\geq 0}$ be monotone and submodular. In the random-order semi-streaming model, the elements of E arrive in a uniformly random permutation, the algorithm uses $r \text{ poly}(\log |E|, 1/\varepsilon)$ space, and it accesses f and M through value and independence oracles. An α -approximation returns a set of expected value at least OPT/α . Offline, continuous greedy attains the value guarantee $1 - 1/e$, and no polynomial-query algorithm can improve it in the value-oracle model, even for a cardinality constraint [16, 58].

Buchbinder, Feldman, Liu, and Sarkar give one-pass algorithms storing

$$O\left(\frac{r}{\varepsilon^2} \log^2 \frac{r}{\varepsilon}\right)$$

elements. Their approximation factors are $1.801 + O(\varepsilon)$ in FPT time and $1.972 + O(\varepsilon)$ in polynomial time [15]. These are the first random-order semi-streaming guarantees strictly better than factor 2. This is a genuine use of random arrival: under adversarial arrival, factor 2 is a lower bound, already for cardinality constraints, for algorithms storing $o(|E|/r^3)$ elements [35]. The target above can be attained with $O(\log(1/\varepsilon))$ passes, but whether one pass suffices remains open.

Open Question 13 (Offline-optimal value in one random-order pass). For every $\varepsilon > 0$, is there a single-pass random-order semi-streaming algorithm that outputs $S \in \mathcal{I}$ satisfying

$$\mathbb{E}[f(S)] \geq \left(1 - \frac{1}{e} - \varepsilon\right) \max_{I \in \mathcal{I}} f(I)?$$

Can such a guarantee be obtained in polynomial time? If not, what are the optimal approximation factors with polynomial and FPT dependence on r ?

14 Monotone submodular maximization over multiple matroids

Background. Let $M_j = (E, \mathcal{I}_j)$, $j \in [k]$, be matroids on the same ground set and let $f : 2^E \rightarrow \mathbb{R}_{\geq 0}$ be monotone and submodular. The problem is

$$\max\{f(S) : S \in \mathcal{I}_1 \cap \dots \cap \mathcal{I}_k\}.$$

A matching in a k -partite, k -uniform hypergraph is already the intersection of k partition matroids, so this framework contains a basic family of packing problems. For $k = 2$, linear objectives are exactly optimizable by matroid intersection, and the common-independent-set polytope is integral. Neither fact makes a nonlinear submodular objective easy: the multilinear extension can have an integrality gap even for bipartite matching.

Write ρ_k for the best worst-case value ratio, so that an algorithm with guarantee ρ_k returns S with $\mathbb{E}[f(S)] \geq \rho_k \text{OPT}$. For every fixed k and $\varepsilon > 0$, Lee, Sviridenko, and Vondrák obtain a $1/(k + \varepsilon)$ guarantee by local search [48]. For two matroids this is usually written $1/2 - \varepsilon$, and no better guarantee is known for two arbitrary matroids. Conversely, no polynomial-query algorithm

can exceed $1 - 1/e$ in general: one matroid may impose a cardinality constraint and all the others may be free [58]. Thus

$$\frac{1}{2} \leq \rho_2 \leq 1 - \frac{1}{e} \approx 0.6321,$$

where the inequalities concern the supremum of achievable guarantees.

For large k , a July 2026 unrefereed preprint of Singer and Thiery claims a $k/2 + o(k)$ loss factor, equivalently $\rho_k \geq 2/k - o(1/k)$ [64]. Its implementation is polynomial for every fixed k , but the exponent depends on k . If the polynomial exponent must be independent of k , Feldman and Ward give an ICALP 2026 algorithm with loss factor

$$\frac{2 \ln 2}{1 + \ln 2} k + O(\sqrt{k}) < 0.819k + O(\sqrt{k}) [37].$$

On the hardness side, combining recent k -CSP hardness with the reduction to multidimensional matching rules out a loss factor smaller than $k/(8 + \varepsilon)$ for every fixed $\varepsilon > 0$ and all sufficiently large k , unless $\text{NP} \subseteq \text{BPP}$ [47, 52]. This already holds for a modular objective and therefore yields the computational upper bound $\rho_k \leq (8 + o(1))/k$.

Open Question 14 (Monotone submodular k -matroid intersection). Determine ρ_2 : can the $1/2 - \varepsilon$ guarantee for two arbitrary matroids be improved, or can the $1 - 1/e$ hardness ceiling be lowered? More generally, close the leading-constant gap

$$\frac{2 - o(1)}{k} \leq \rho_k \leq \frac{8 + o(1)}{k},$$

and determine whether the $k/2 + o(k)$ loss factor can be achieved in time whose polynomial exponent is independent of k .

15 Exact and approximate submodular partitioning

Background. Let $f : 2^V \rightarrow \mathbb{R}_{\geq 0}$ be a submodular function given by a value oracle, where $N = |V|$. In *submodular k -partition*, the goal is to partition V into k nonempty sets while minimizing

$$\min \left\{ \sum_{i=1}^k f(V_i) : V = V_1 \dot{\cup} \dots \dot{\cup} V_k, V_i \neq \emptyset \right\}.$$

The problem is polynomial-time solvable for $k = 2$ by submodular-function minimization, for $k = 3$ by Okumoto, Fukunaga, and Nagamochi, and for $k = 4$ by Hirayama, Liu, Makino, Shi, and Xu [59, 42]. Thus $k = 5$ is the first unresolved fixed value.

Several nearby graph and hypergraph problems clarify the boundary. For a hypergraph cut function d_H , the objective $\sum_i d_H(V_i)$ is called *Hypergraph- k -Partition*: a hyperedge meeting $h \geq 2$ parts is charged h times. It is different from *Hypergraph- k -Cut*, which charges each crossing hyperedge only once. Hypergraph- k -Cut has a deterministic $N^{O(k)}$ -time algorithm for every fixed k [19], while Hypergraph- k -Partition remains open already for fixed $k = 5$. For ordinary graphs every crossing edge meets exactly two parts, so minimizing the symmetric cut-function objective is equivalent, up to a factor of two, to Graph- k -Cut. Although the two hypergraph objectives differ, Hypergraph- k -Cut can itself be encoded as a submodular k -partition instance [21].

When k is part of the input, a greedy splitting algorithm gives a $(k - 1)$ -approximation for arbitrary nonnegative submodular functions [69]. On the other hand, the Hypergraph- k -Cut special case rules out an $N^{1/(\log \log N)^c}$ -approximation for some universal $c > 0$ under the Exponential

Time Hypothesis [21]. For broad structured classes the value-oracle thresholds are much sharper. The principal-partition-sequence algorithm gives a $4/3 - 4/(9N + 3)$ approximation for monotone functions, matching the asymptotic $4/3$ value-oracle lower bound; for symmetric and posimodular functions it gives factor 2, which is also asymptotically optimal in the value-oracle model [62, 20].

Two elementary monotone special cases still lack matching computational bounds. If $f(S) = r_M(S)$, the objective minimizes the sum of the ranks of the parts; for a represented matroid, this partitions the columns of a matrix to minimize the sum of the dimensions of their spans. If $b(S)$ is the weight of graph edges incident with S , then b is a coverage function and, for a partition $\mathcal{P}$,

$$\sum_{P \in \mathcal{P}} b(P) = w(E) + w(\delta(\mathcal{P})).$$

Thus graph-coverage partition and Graph- k -Cut have the same optimal partition, although their approximation ratios are not equivalent because of the additive term $w(E)$. The former is NP-hard and has a $4/3$ -approximation, but its precise approximability, like that of matroid-rank partition, remains open [62, 20].

A second central variant prescribes terminals $t_1, \dots, t_k$ and requires $t_i \in V_i$. For monotone submodular multiway partition, the best general factor is $4/3$, while a factor below $10/9$ requires exponentially many value queries. For graph coverage, the corresponding bounds are $9/8$ and 1.00036 under $P \neq NP$ [12].

Open Question 15 (Submodular partition frontiers). Resolve the following fundamental gaps.

- (a) Is submodular k -partition polynomial-time solvable for every fixed k ? In particular, solve $k = 5$, even for symmetric functions or explicitly given hypergraph cut functions.
- (b) Determine the approximability of arbitrary nonnegative submodular k -partition as a function of N and k , between the current $(k - 1)$ guarantee and the ETH-based hardness inherited from Hypergraph- k -Cut.
- (c) Can the $4/3$ guarantee be improved for matroid-rank or graph-coverage k -partition, or can matching computational lower bounds be proved?
- (d) Close the gaps $[10/9, 4/3]$ for monotone submodular multiway partition in the value-oracle model and $[1.00036, 9/8]$ for its graph-coverage special case under $P \neq NP$.

16 Optimal monotone contention resolution for matroid intersections

Background. Contention resolution isolates the rounding part of a relaxation-based algorithm. Let

$$P = \bigcap_{j=1}^k P(M_j)$$

be the intersection of the independence polytopes of k matroids. Given $x \in bP$, form $R(x)$ by activating each element e independently with probability x_e . A (b, c) -balanced contention-resolution (CR) scheme returns a feasible subset $\pi_x(R(x)) \subseteq R(x)$ such that

$$\Pr[e \in \pi_x(R(x)) \mid e \in R(x)] \geq c \quad \text{whenever } x_e > 0.$$

It is *monotone* if an element's retention probability cannot increase when more competing elements are presented. This property refers to the rounding rule, not to whether the objective function is monotone, and it is what lets the marginal guarantee transfer cleanly to submodular value.

Put $q(b) = (1 - e^{-b})/b$. Chekuri, Vondrák, and Zenklusen construct an optimal monotone $(b, q(b))$ scheme for one matroid and show that monotone schemes compose multiplicatively [22]. Applying the single-matroid scheme separately to all k constraints therefore gives

$$(b, q(b)^k) = \left(b, \left(\frac{1 - e^{-b}}{b}\right)^k\right).$$

This is the best general monotone guarantee currently known, but optimality of each one-matroid factor does not imply optimality of their composition.

Let $C_k^{\text{mon}}(b)$ be the largest balance guaranteed for every intersection of k matroids. Bipartite matching is the intersection of two partition matroids. Bruggmann and Zenklusen determine its optimal monotone balance

$$\beta(b) = \mathbb{E} \left[\frac{1}{1 + \max\{X_1, X_2\}} \right], \quad X_1, X_2 \text{ independent Pois}(b) [13].$$

It follows that

$$q(b)^2 \leq C_2^{\text{mon}}(b) \leq \beta(b),$$

and in particular

$$0.3995764009 = (1 - 1/e)^2 \leq C_2^{\text{mon}}(1) \leq 0.4762223882.$$

The upper endpoint is achieved for bipartite matchings; it is only an upper bound, not a construction, for arbitrary pairs of matroids. Without the monotonicity requirement, Adamczyk and Włodarczyk give a random-order online $(1, 1/(k+1))$ scheme [2]; this different model does not settle the monotone offline question.

Open Question 16 (Best monotone CR scheme for k matroids). Determine $C_k^{\text{mon}}(b)$. Already for $k = 2$ and $b = 1$, can a joint scheme for two arbitrary matroids improve on independent composition, perhaps up to the bipartite-matching ceiling, or can a sharper obstruction be proved? More generally, what is the dependence on k and what is the precise price of requiring monotonicity?

17 Online and random-order contention resolution

Background. An offline CR scheme sees the entire active set before resolving conflicts. An online contention-resolution scheme (OCRS) instead learns, one element at a time, whether that element is active and must accept or reject it irrevocably while maintaining feasibility [36]. The order model is essential. A *fixed-order* adversary commits to the order before seeing the active set or the algorithm's randomness; an *almighty* adversary sees both; and in a random-order CR scheme (ROCRS), the order is a uniformly random permutation.

For independent activations with marginal vector in a single matroid polytope, Lee and Singla give an optimal $1/2$ -balanced OCRS in the fixed-order model and an optimal $(1 - 1/e)$ -balanced ROCR [46]. Against an almighty adversary, the best general matroid construction guarantees $1/4$ after optimizing the standard scaling, while $1/2$ remains an upper bound [36, 46]. Chapter 12's broad question about cardinality constraints now needs qualification: Dinev and Weinberg determine the asymptotic behavior for uniform matroids in the fixed-order and almighty models, while exact finite-rank constants and general matroids remain open [27].

Random order becomes still more subtle when activations may be correlated. A distribution μ on subsets of E is α -*uncontentious* if some offline map can retain every element with probability at least α times its activation probability. A *universal* ROCRS would preserve a constant fraction of this best offline balance for every such distribution. Dughmi proves that the existence of a constant-factor universal ROCRS for all matroids is equivalent, up to constants, to the matroid secretary conjecture [29].

For intersections, Adamczyk and Włodarczyk obtain a not-necessarily-monotone $1/(k+1)$ -balanced ROCRS for k matroids [2]. Even the $k=2$ special case of bipartite matching is unresolved: the best random-order balance lies between 0.4789 and $1/2$ [50]. These results concern marginal balance; for nonmonotone submodular objectives, a better balance is useful only if the scheme also has enough correlation structure to preserve expected submodular value.

Open Question 17 (Fundamental OCRS and ROCRS frontiers). Resolve the following basic online contention-resolution questions.

- (a) Determine the optimal balance for a general matroid against an almighty adversary, closing the current gap between $1/4$ and $1/2$.
- (b) Does every matroid admit a constant-factor universal ROCRS for all correlated distributions that are already constant-balanced offline? Equivalently, resolve the matroid secretary conjecture.
- (c) Determine the optimal ROCRS balance for intersections of k matroids, including the interval $[0.4789, 1/2]$ for bipartite matching when $k=2$, and identify when such schemes can also round nonmonotone submodular objectives without an additional loss.

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